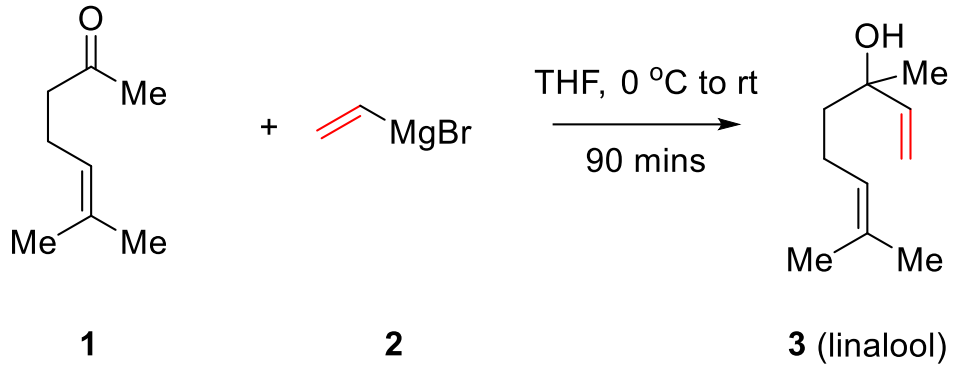
CM3191 Org L4: Schlenk line techniques

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Synthesis of linalool

- Reaction scheme



1 **2** **3** (linalool)



Reagent	MW	mmol	amount
6-methyl-5-hepten-2-one	126.20	3.0	0.44 mL
vinylmagnesium bromide 1M solution in THF	131.25	7.5	7.5 mL
THF	72.11	NA	5 mL
Product	154.25	___	___

Nitrogen atmosphere



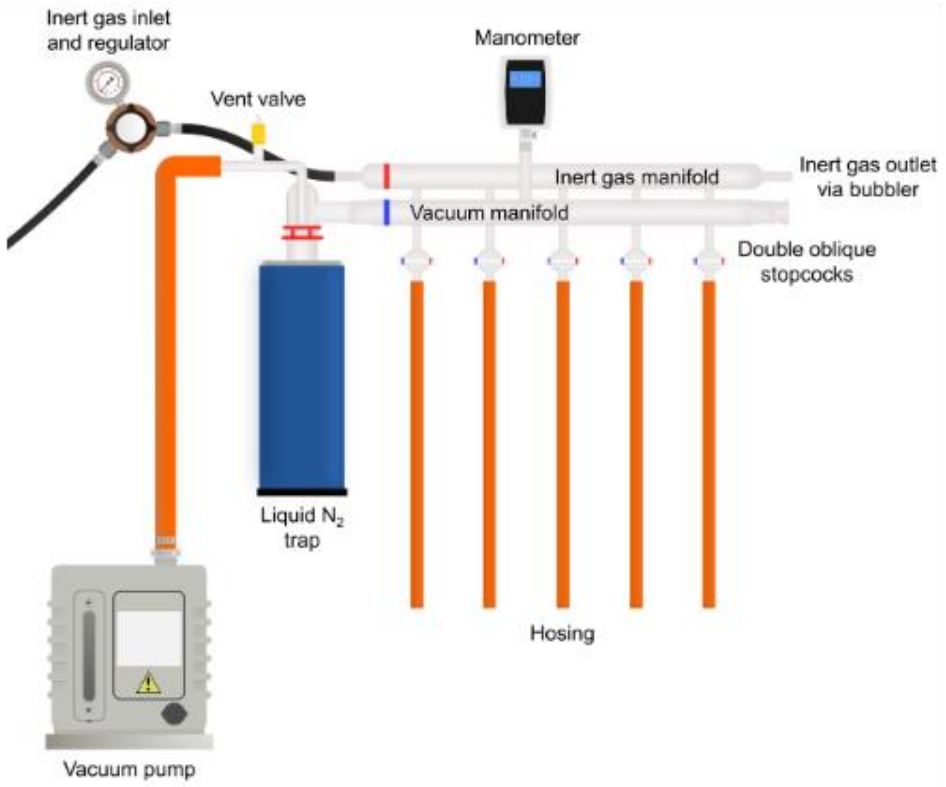
Grignard bottle under a nitrogen balloon



A reaction under nitrogen balloons

Schlenk Line

- Purpose: A Schlenk line enables the safe manipulation of air- and moisture-sensitive compounds.



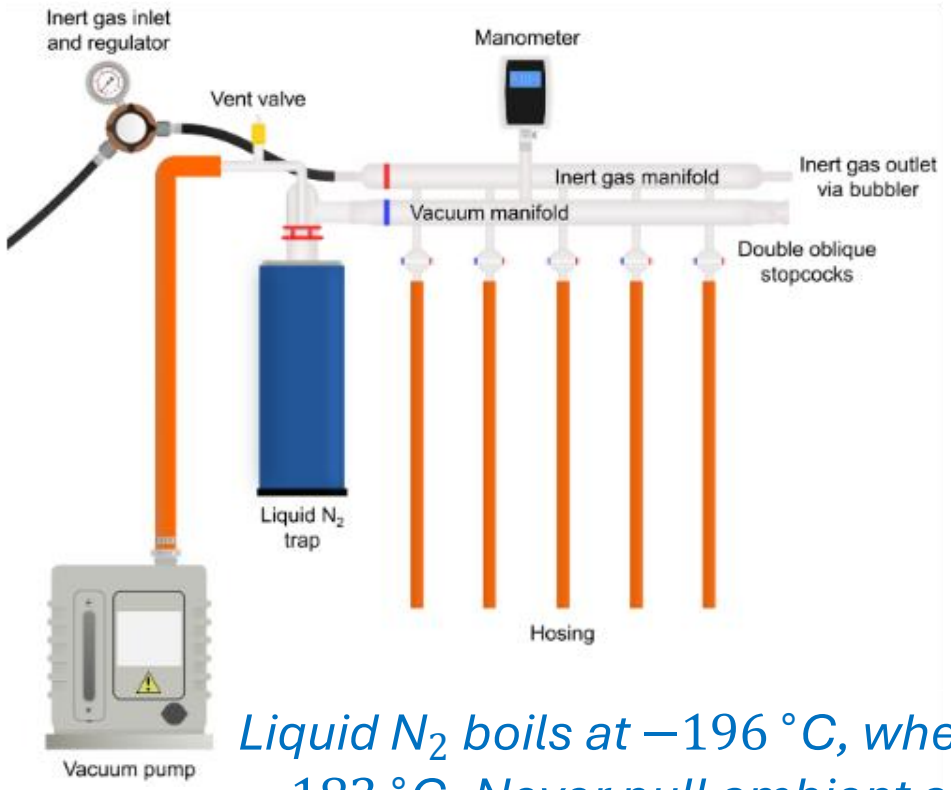
A Schlenk line is a dual-manifold system:

- Inert gas manifold: Connected to a source of purified and dry inert gas (nitrogen or argon)

- Vacuum manifold: Connected to a high-vacuum pump, typically reaching operating pressures of 10^{-2} to 10^{-4} mbar

https://schlenklinesurvivalguide.wordpress.com/

Bubbler and Cryogenic trap



Bubbler (mineral or silicone oil): Acts as a pressure-relief system, provides a visual monitor of gas flow rate, and prevents air/moisture ingress into the manifold.

Cryogenic trap: Condenses volatile vapors and solvents prior to the pump, protecting the vacuum pump mechanism and oil from contamination.



*Liquid N_2 boils at −196 °C, whereas O_2 condenses at − 183 °C. Never pull ambient air through a liquid-N_2 trap; condensing atmospheric O_2 into organic solvent residue forms **severely explosive liquid oxygen mixtures**.*

Alternative (safer) cold trap: Dry ice / acetone bath (−78 °C).

https://schlenklinesurvivalguide.wordpress.com/


Vacuu
Nitro n gas

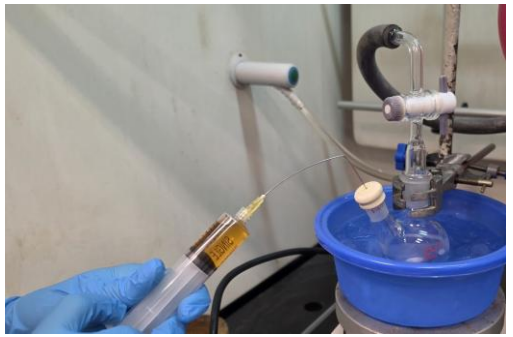


Image source: Sigma-Aldrich

The line features multiple ports fitted with double-oblique stopcocks (glass or PTFE/Teflon), allowing connected vessels (Schlenk flasks or Schlenk tubes) to be independently evacuated or backfilled with inert gas.

Liquid Transfer: Achieved via inert-gas purged **syringe & needle** ($\leq$ 20 mL) or via **cannula transfer** for larger volumes.

Syringe Preparation: Always cycle/purge syringes with inert gas 3 times prior to drawing sensitive reagents.



Connecting (Cycling) glassware onto the Schlenk Line

- Setup & Evacuation: Assemble the glassware (dried overnight in an oven) or connect it and apply heat with a heat gun under vacuum.
- Evacuation Period: Evacuate under vacuum while allowing the flask to cool completely to room temperature (~5 min for a 100 mL vessel).
- Backfill: Slowly turn the stopcock to backfill the vessel with inert gas.
- Repeat: Complete three full vacuum–refill cycles to ensure thorough removal of air and surface moisture.

Caution: Preventing Bubbler Suck-Back

Opening an evacuated flask too quickly draws inert gas faster than the supply can replenish, pulling bubbler oil into the line.

→ Confirm positive inert gas flow and backfill the vessel (VERY) slowly.



Readings

1. A. M. Borys. “An Illustrated Guide to Schlenk Line
2. Techniques” *Organometallics*, **2023**, *42* (3), 182-196.
   A. M. Borys. “The Schlenk Line Survival Guide” https://schlenklinesurvivalguide.com